

MethaNet Molecular attestation layer



A graph that connects candidates to evidence, provenance, claim status, and validation paths

Multi-source molecular signals

ESM-2 geometry

662 proteomes; bridge-neighbor signal

gLM2 context

gene order and neighborhood features

Functional atlas

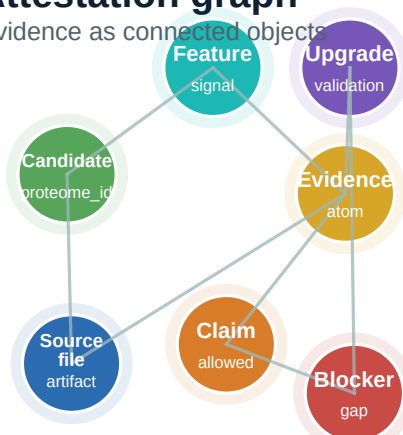
methane, sulfur, substrate annotations

QC + provenance

taxonomy, completeness, files, checksums

Attestation graph

Evidence as connected objects



candidate -> evidence -> source file
claim -> gap -> validation

Claim-aware outputs

Why it matters

candidate + function + context

Evidence supports it

geometry + annotation + QC

File produced it

manifest path + checksum

Claim allowed

attestation, not final MRV

Claim blocked

abundance / environment / validation missing

Upgrade action

sample mapping + coverage + field data

Claim boundary

MAG/proteome molecular screening now; sample-level risk needs abundance, environmental context, uncertainty, and flux/process validation.